

THE DUAL-CENTER COMMUTANT PROBLEM ON UNCOUNTABLE BLOCK SUMS

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STATUS AND SOURCE QUESTION

This packet gives a partial positive answer to the open problem in Remark 4 of Orhon [3]. If $i : Z(E) \rightarrow \mathcal{L}(E)$ is the natural embedding and E has a topologically full center, the question is:

$$T \in \mathcal{L}(E'), \quad Ti^*(a) = i^*(a)T \quad (a \in Z(E)) \implies T \in Z(E') ?$$

The cited book [1, §9.11(1)] proves this when E has a quasi-interior point. The theorem below treats arbitrary-index c_0 - and ℓ_p -sums of such lattices. For an uncountable index set, the resulting lattice generally has no quasi-interior point.

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topologically full center.

Remark 4. When E is a Banach lattice with topological order unit a much stronger result than Corollary 3 is proved in ([1], 9.11, (1)): *Let T be an operator on E' that commutes with $i^*(Z(E))$, then T is in $Z(E')$.* It is further stated in [1] that the result remains true if E is a Banach lattice with topologically full center. However this latter statement is still an open problem. For a more general version of this problem, the interested reader should see ([7], 3., p. 357). It is worth noting that the other two questions raised in ([7], 3.) have been answered affirmatively by Arenson's Theorem.

Remark 5. Let E be a Riesz space with point separating order dual $E^\sim$. We will say

DEFINITIONS AND NOTATION

The ideal center of a Banach lattice F is

$$Z(F) = \{S \in \mathcal{L}(F) : |S| \leq CI_F \text{ for some } C \geq 0\}.$$

Equivalently, it is the algebra of bounded orthomorphisms dominated in modulus by a scalar multiple of the identity.

Let Γ be any index set and let $(E_\gamma)_{\gamma \in \Gamma}$ be Banach lattices. We use the coordinatewise lattice structure on

$$E = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_p \quad (1 \leq p < \infty), \quad \text{or} \quad E = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_{c_0}.$$

In the c_0 case, $\|x_\gamma\| \rightarrow 0$ means that for every $\varepsilon > 0$ only finitely many coordinates have norm at least ε .

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IDEA OF THE PROOF

The coordinate projections belong to $Z(E)$. Consequently any operator on E' commuting with the adjoint copy of $Z(E)$ must be block diagonal. Its γ th block commutes with all adjoints of $Z(E_\gamma)$, because a component-center operator can be extended by zero to a member of $Z(E)$. The known quasi-interior theorem therefore makes every block central. The block norms are uniformly bounded by the norm of the original operator, so the blocks reassemble into a member of $Z(E')$.

The same finite-coordinate approximation also proves directly that the primal block sum has a topologically full center.

PARTIAL THEOREM

Theorem 1. *Let Γ be an arbitrary index set. Suppose that every Banach lattice E_γ has a quasi-interior point, and let E be either*

$$\left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_p \quad \text{for some } 1 \leq p < \infty, \quad \text{or} \quad \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_{c_0}.$$

Then E has a topologically full center. Moreover, if $T \in \mathcal{L}(E')$ commutes with $i^(Z(E))$, then $T \in Z(E')$.*

Proof. For each γ , let Q_γ be the coordinate band projection on E . The standard coordinate argument gives the isometric algebra and lattice identification

$$(1) \quad Z(E) = \left\{ (S_\gamma)_{\gamma \in \Gamma} : S_\gamma \in Z(E_\gamma), \sup_\gamma \|S_\gamma\| < \infty \right\},$$

where the family acts coordinatewise. Indeed, a central operator preserves the bands $Q_\gamma E$ and its restrictions are central; the converse follows immediately from $|S_\gamma| \leq \|S_\gamma\| I_{E_\gamma}$.

We first verify topological fullness. Fix $x = (x_\gamma) \in E_+$ and put

$$J_\gamma = \overline{Z(E_\gamma)x_\gamma} \subset E_\gamma.$$

Because E_γ has a quasi-interior point, its center is topologically full [3], so J_γ is an ideal. We claim that

$$(2) \quad \overline{Z(E)x} = \left(\bigoplus_{\gamma \in \Gamma} J_\gamma \right)_p \quad \text{or} \quad \left(\bigoplus_{\gamma \in \Gamma} J_\gamma \right)_{c_0},$$

respectively. One inclusion follows from (1). For the other, take y in the right-hand side and $\varepsilon > 0$. Truncate y to a finite set $F \subset \Gamma$ with error below $\varepsilon/2$. For every $\gamma \in F$, choose $S_\gamma \in Z(E_\gamma)$ so that $S_\gamma x_\gamma$ approximates y_γ closely enough that the finite-sum error is below $\varepsilon/2$. Extending this finite family by zero gives an element of $Z(E)$ and proves (2). Its right-hand side is an ideal, proving that E has a topologically full center.

Now suppose that $T \in \mathcal{L}(E')$ commutes with $i^*(Z(E))$. Since $Q_\gamma \in Z(E)$, we have

$$TQ'_\gamma = Q'_\gamma T \quad (\gamma \in \Gamma).$$

Thus T leaves $Q'_\gamma E' \cong E'_\gamma$ invariant; denote the restriction by T_γ . If $S \in Z(E_\gamma)$, extend S by zero on all other coordinates. Formula (1) puts the extension in $Z(E)$, so T_γ commutes with the natural adjoint copy of $Z(E_\gamma)$. The quasi-interior theorem [1, §9.11(1)] yields

$$(3) \quad T_\gamma \in Z(E'_\gamma) \quad \text{and} \quad \|T_\gamma\| \leq \|T\|.$$

Let q be conjugate to p . The dual is

$$E' = \left(\bigoplus_{\gamma \in \Gamma} E'_\gamma \right)_q$$

with the usual interpretation $q = \infty$ when $p = 1$; in the c_0 case it is the ℓ_1 -sum. For every $x' \in E'$ and every γ ,

$$(Tx')_\gamma = Q'_\gamma Tx' = TQ'_\gamma x' = T_\gamma x'_\gamma.$$

This coordinate identity also handles $q = \infty$: an element of the product is determined by all its coordinates, so there is no additional residual operator. By (3), the uniformly bounded family (T_γ) defines a coordinatewise central operator on E' , dominated in modulus by $\|T\|I_{E'}$. It agrees with T on every coordinate and hence equals T . Therefore $T \in Z(E')$. $\square$

WHY THE FULL PROBLEM REMAINS

Orhon's topological-fullness theorem identifies $Z(E') = i^*(Z(E)'')$ through an Arens extension. Approximating an element of $Z(E')$ by elements of $i^*(Z(E))$ gives only weak-star operator convergence. An arbitrary $T \in \mathcal{L}(E')$ need not be weak-star continuous, so its commutation cannot be passed through that limit. The block-sum proof avoids this by using actual coordinate projections already in $Z(E)$.

The method does not automatically cover ℓ_∞ -sums. Their Banach duals contain a singular part beyond the coordinate dual bands, and the coordinate equation no longer exhausts the dual. Controlling or exploiting that singular part is a plausible route to the unrestricted problem.

VERIFICATION AND NOVELTY NOTES

The proof is structural and uses no numerical computation. The primal-center identification, topological-fullness argument, dual coordinate identity, and $p = 1$ product case were checked separately. The local run indexes contained no duplicate. A local source search found Alpay–Orhon [2], which proves a related theorem for adjoints of order-bounded primal operators; that extra adjoint continuity does not apply to arbitrary operators on E' . Bounded exact-phrase web searches on 2026-08-11 found no direct later resolution, but some final targeted calls timed out, so novelty confidence is moderate. Human review should focus on the standard center identifications and the cited quasi-interior theorem.

REFERENCES

- [1] Y. A. Abramovich, E. L. Arenson, and A. K. Kitover, *Banach $C(K)$ -modules and operators preserving disjointness*, Pitman Research Notes in Mathematics Series 277, Wiley, 1992.
- [2] S. Alpay and M. Orhon, *Characterization of Riesz spaces with topologically full center*, Positivity 19 (2015), 799–816; arXiv:1406.6335.
- [3] M. Orhon, *The ideal center of the dual of a Banach lattice*, Positivity 15 (2011), 555–564; arXiv:1002.4346.